

My research focuses on galaxy evolution, with particular emphasis on the impact of AGN feedback and galaxy mergers on star formation, in sub-kpc scale via gravitationally lensed galaxies. AGN feedback is widely considered a key regulator of star formation and gas accretion (Ferrarese & Ford 2005; Churazov et al. 2005; Springel et al. 2005; McNamara & Nulsen 2007; Cattaneo et al. 2009; Fabian 2012; Kormendy & Ho 2013), while galaxy merger is known to trigger enhanced star formation/starbursts and contribute significantly to mass assembly, particularly around the peak epoch of galaxy growth at Cosmic Noon (Toomre & Toomre 1972; Barnes & Hernquist 1996; Mihos & Hernquist 1996; Hopkins et al. 2006; Conselice 2014).

I plan to study the AGN feedback and galaxy mergers with larger samples across the cosmic time, as well as the spatially decomposed source plane reconstruction with sub-kpc scales for lensed SFGs/starbursts in the early Universe, to explore the understanding of the galaxy evolution.

Research Experience & Key Contributions

During my PhD at Nanjing University, I mainly investigated the impact of AGN jet feedback and galaxy merger on the evolution of the Cloverleaf quasar (H1413+117), a gravitationally lensed starburst galaxy at the redshift of 2.558, in a spatially resolved ~ 500 -pc scale.

Discovery of a Lensed Single-Sided Jet: Utilizing the lens mass model of the Cloverleaf quasar (from synthesis ALMA observations at 300 GHz) and non-parametric source-plane reconstruction with **PyAutoLens**, I analyzed radio continuum emission from e-Merlin (L-band) and VLA (X & Ka bands) and revealed a single-sided jet located 1 kpc from the dusty disk (reconstructed from observing-frame 300 GHz continuum). SED fitting in the source plane confirmed that radio emission from the disk is dominated by star formation (Zhang et al. 2023, Zhang et al. submitted), whereas emission at the jet position is dominated by synchrotron radiation. This work suggests that AGN feedback in the Cloverleaf quasar is driven by both AGN winds and jets.

Jet-ISM Interaction and Quenching: Using multiple SFR tracers ($\text{H}_2\text{O } 4_{22} - 4_{13}$, rest-frame 120 μm , and 36 GHz continuum) and CO(3-2) as a cold interstellar-medium (ISM) tracer, I employed non-parametric modeling to investigate AGN feedback effects. I found that the star formation efficiency (SFE) on the jet-hosting side of the disk is suppressed by 30% compared to the counter side. Kinematic modeling with ^{3D}BAROLO indicates that the jet interacts with gas along its path, actively suppressing SFE. Based on the proposed observation conducted by VLBA (project ID: BZ105), with a beam size of $0''.03$, two knot-structure radio components at L-band near the caustics with the spatial resolution of ~ 30 pc, indicating tens-pc scale jet-ISM interaction. This suggests that direct localized jet-ISM interaction serves as a potentially major quenching channel during the rapid quenching phase in radio-quiet quasars (Zhang et al. submitted).

Merger-Driven CGM Growth: By combining CO(3-2) data of the Cloverleaf from various configurations, I identified two molecular tidal tails with the half of the dynamical disk mass, extending tens of kpc in the CGM, indicative of an interaction with a neighbor galaxy. Non-parametric lens modeling revealed the root of a tidal tail stripping molecular gas from the Cloverleaf disk. This suggests that major mergers at Cosmic Noon can remove up to half of the molecular cold ISM into the CGM, significantly enriching it in the early encounter phase, which is hardly expected in observations (Zhang et al. in preparation). This work indicates that cold gas removal through tidal tails in mergers could suppress star formation in a more dramatic way than previously expected.

Future Research Interests & Plans

Hundreds-pc-scale view of molecular excitation and heating in lensed high- z galaxies

Strong lensing is a promising route to sub-kpc molecular-gas physics at high redshift, but its full power remains unlocked: most lensed CO studies still fit galaxy-integrated line ratios, folding differential magnification, multi-phase gas, and mixed heating into a single set of numbers. This program breaks this degeneracy by moving the entire excitation analysis into the source plane, pixel by pixel, and carrying it through to a spatially resolved attribution of heating mechanisms. The workflow lists below, each feeding the next.

- **Source-plane spectral data cube.** I will reconstruct multi-transition molecular maps (CO ladder, isotopologues, dense-gas tracers, CI) into a dynamic-resolution, non-parametric, and pixelized source-plane grid. The spatial distribution and the kinematics of the individual gas components will be recovered. Differential magnification is measured per transition from the lens model, not assumed away.
- **Pixelized physical conditions.** Each source-plane pixel is fit independently with a non-LTE escape-probability solver (RADEX-class), with LTE rotation diagrams as a baseline and cross-check. The CO ladder constrains kinetic temperature, high-critical-density lines (HCN, HCO⁺, HNC) constrain volume density, and optically thin anchors (¹³CO, C¹⁸O, CI, dual-band dust continuum) break the column-density degeneracy. The deliverable is a set of source-plane maps, T_{kin} , n_{H_2} , N_{mol} , with full per-pixel posteriors, so that uncertainty, not just best-fit values, is spatially resolved.
- **Pixelized heating attribution.** The measured line intensities are fitted per-pixel against PDR (PDR Toolbox, Cloudy, KOSMA-tau), XDR (Cloudy XDR), and shock (Paris-Durham) grids, with the RADEX-derived conditions serving as consistency priors, yielding maps of FUV field, X-ray deposition, and shock velocity.
- **AGN & star formation energy budget constrain in observations.** Line-ratio fits alone are degenerate, so every pixel also passes an energy-budget check: star formation energy budget constrained by FUV from the resolved SFR surface density (inferred from FIR, free-free, and high-lying water emission), AGN energy budget constrained by X-ray and mechanical input from Chandra/XMM and radio jet/outflow data, against the observed molecular and fine-structure cooling. Heating is claimed only where both the grid fit and the budget close, anchored by shock tracers and outflow kinematics where shocks are invoked.

The payoff is a new observable class: heating maps of galaxies at high redshifts, showing where star formation, AGN, and shocks each sustain the molecular phase. This turns lensed galaxies from bright but blurred laboratories into spatially resolved testbeds for feedback physics, directly comparable to simulations with explicit stellar and AGN feedback channels.

New quenching scenario in galaxy evolution

As I have found two potentially under-estimated quenching channels (i.e. jet feedback in radio quiet quasars and cold ISM removal via tidal tails during mergers) in the unique laboratory, the Cloverleaf. To improve our understanding of the quenching scenario and the galaxy evolution, I

aim to systematically study the following projects on larger samples.

- **Localized AGN-jet feedback on star formation in SFGs:** About 40% of radio-quiet quasars are found to contain jet (Njeri et al. 2026), but the jet feedback on star formation is overlooked. I plan to study lensed SFGs which show radio flux exceeding star formation activity, by the high-spatial/spectral-resolution data of SFR tracers and radio emission, with observations from JWST/ALMA/VLA/VLBI, to analyze the localized AGN-jet feedback on star formations and re-estimate the relative role of multiple feedback channels, as well as their evolution across cosmic time, based on non-parametric source plane reconstruction in hundreds-of-pc scale. With the observation by EAVN, VLBA, and other VLBI facilities, tens-of-pc scale imaging could be available for detailed jet feedback and its impact on star formation.
- **Gas transfer in merger events:** Merger rate is higher around Cosmic Noon, but at which redshift the spatial resolution is limited to galaxy scale. By source reconstruction in lensed merging systems, I aim to study the merger-driven structures in both ISM-scale and CGM-scale. This helps reveal the obscured gas transfer process during a merger, as well as the star formation fuel removal which could decrease the final total stellar mass.

With the projects above, I hope to answer these questions:

Is jet-ISM interaction a popular star formation suppression channel in the early universe?

How does localized jet impact on star formation regions?

Is merger-driven gas removal a significant way to suppress the star formation in the rapid quenching phase?

How do physical conditions differ and what mechanism leads to the differences, in starbursts with sub-kpc scale?

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